JOHN JAY COLLEGE OF CRIMINAL JUSTICE
The City University of New York
524 West 59th Street, New York, NY, 10019

Syllabus for: MAT/FOS 705
Applied Statistics and Data Analysis for Forensic Science and Forensic Computing

Professor: Nicholas Petraco
Room: NA

Contact hours: Any time. Just email me at the address below.
Course Format: Asynchronous instruction with optional live Zoom tutorials and review sessions to be announced throughout the semester.
E-mail address: npetraco@gmail.com
Course website: https://npetraco.github.io/MATFOS705/

Course Description:
The classical "work horse" statistical methods are known as "frequentist" and "Fisherian" methods. They are extremely useful and have been successfully applied for almost one hundred years. The purpose of this hands-on course is to acquaint forensic computing and forensic science graduate students with statistical methods that are applicable to evidence they will encounter, as well as issues and pitfalls to be aware of when applying these methods.

It is not adequate to simply learn the "theory" behind the methodology encountered in this course. It must be applied on real data of practical interest to the forensic science. As such this course will build expertise in the general scientific/statistical computing environment R (http://www.r-project.org/). The course assumes minimal knowledge of numerical computing and statistical procedures. It is designed to build the student's skill set and confidence in both of these areas. Topics covered will include basic data descriptive tools, graphing, probability theory, discrete/continuous distributions, estimation, hypothesis testing and regression.

This is an asynchronous instruction-based course. This means that the entire course is on-line and ***there is no officially scheduled or required live meeting time***. Course lecture materials will be posted on the following website:

https://npetraco.github.io/MATFOS705/

Announcements and important reminders will be emailed to you. As such **you must give me an email address that you check on a regular basis**. It will be our only official method of communication.

Most lectures posted to the course website will be accompanied by short videos which will be posted to YouTube. Links to the videos will also be sent to you in email when they are posted. Periodic live tutorial and review sessions over Zoom will be scheduled throughout the semester

in email. They are not mandatory and will be recorded. After they have occurred, links to the recordings will be sent out to the class. Homework will be given though WebAssign. See below for details.

Course Learning Goals:

1. Recognize the importance of accuracy and objectivity in collecting/sampling data for applications to the law and policy.

2. Acquire an understanding of the types of data that can be recorded and analyzed (e.g. for quantitative trace, fire debris, toolmark, spectrochemical and digital evidence analysis).

3. Acquire an understanding of frequentist statistical tools that can be used to analyze collected forensic data.

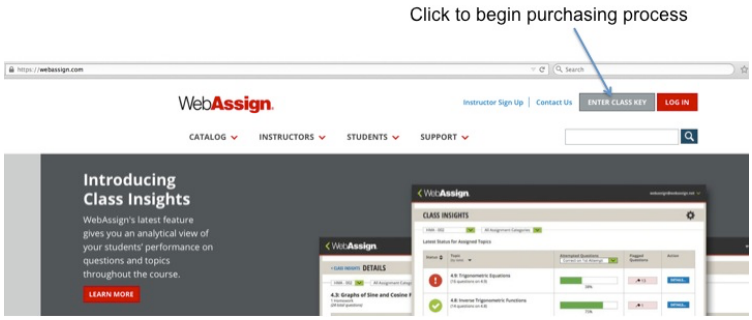
4. Understand the limitations of the frequentist statistical methods used for data analytics and how not to misrepresent the capabilities of these methods to the courts or clients (ethics).

5. Obtain skill with the general computing/statistical software **R** (http://www.r-project.org/).

6. Develop oral and written communication skill as to how to present the results of sophisticated quantitative analysis to officers of the course and lay juries, in terms that are understandable to them.

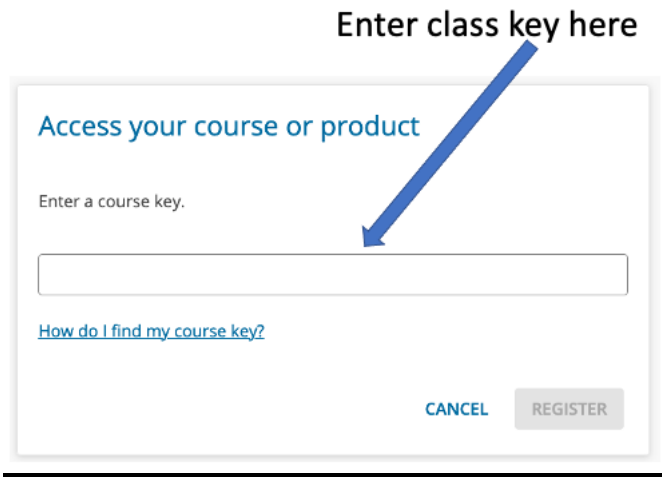
Required Electronic Resources:

- **Webassign**: Can be purchased at:
    - **https://webassign.com/**
    - In order to purchase click on "**Enter Class Key**":

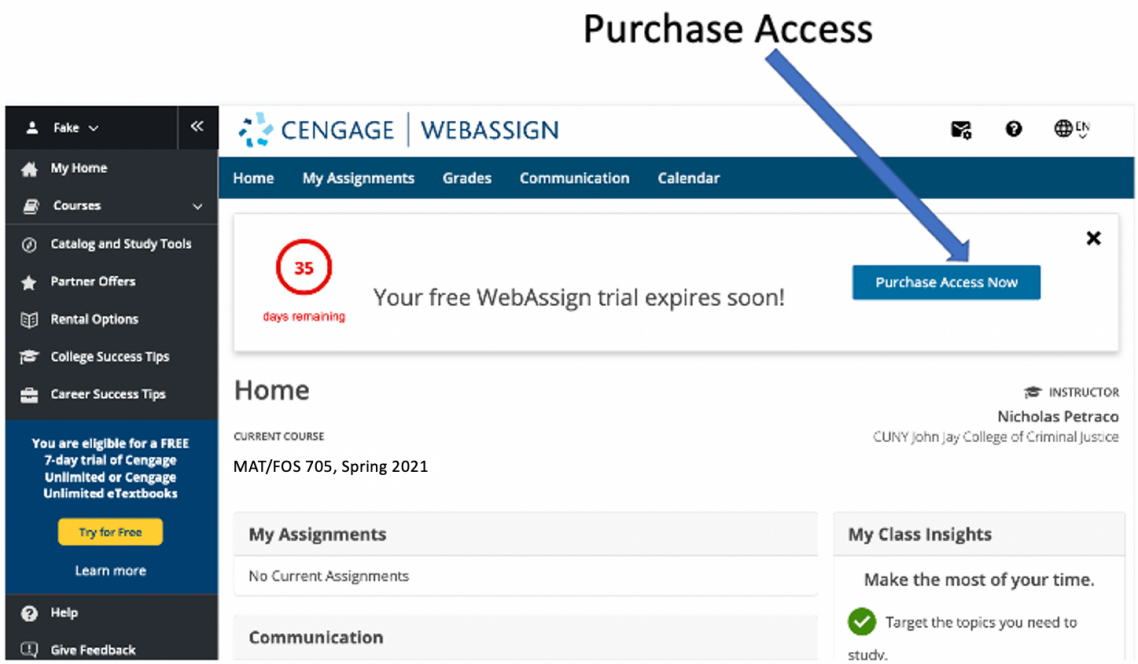


- The WebAssign website will probably prompt you to log into your account. If you don't have an account, create one.
- When you log in, you should see a place to enter the class key:

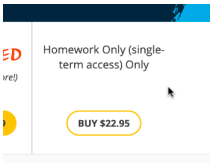
- Class Key: **jjay.cuny 7307 3187**



- After logging in/creating-account, select your class and the website will prompt you to purchase the required materials:



- Purchase "MAT/FOS 705", which should be ~$22.95:



Suggested Textbooks:
The R Book
Crawley
ISBN-10: 8126569719

Introduction to Data Analysis with R for Forensic Scientists
Curran
ISBN-10: 1420088262

Grading:

The grades for this course are based on homework (100%). We'll have about HW set one week.

Topics and Rough Schedule:

Week	Lecture Topics	Sub Topics	HW Due Dates
Feb 1-5	Introduction	Introduction to Bayesian Perspectives Review: Common Defininitions in Statistics Review: Basic Graphing	
Feb 8-12	Introduction and Tutorial for R		Feb 12: HW Set 1 Due
Feb 15-19	Review: Summary Statistics 1-3	Review: Measurs of Central Tendency Review: Standard Measures of Uncertainty Review: Range and Quantiles	Feb 22: HW Set 2 Due
Feb 22-26	Probability 1-3	Definitions Axioms and Some Useful Theorems Conditional Probability and Bayes' Theorem	Mar 1: HW Set 3 Due
March 1-5	Important Distributions 1-2	Important Discrete Distributions Important Continuous Distributions	Mar 8: HW Set 4 Due
March 8-12	Review		
March 15-19	Catch up		
March 22-26	Discrete Moments and Peobability Mass	Permutations and Combinations Discrete Probability Mass Discrete Moments	Mar 29: HW Set 5 Due
Mar 29-Apr 2	Point Estimation 1-2	Maximum Likelihood Estimators Sampling Distributions	Apr 5 HW Set 6 Due
April 5-9	Point Estimation 3-4	Unbiased Estimators Example Problems	Apr 12: HW Set 7 Due
April 12-16	Point Estimation 4-6	Bootstrap Estimation Bootstrap Fails	Apr 19: HW Set 8 Due
April 19-23	Interval Estimation 1-2	Definitions and Theory Computing Confidence Intervals	Apr 26: HW Set 9 Due
April 26-30	Interval Estimation 3-4	Reference Formulas Bootstrap Confidence Intervals	May 3: HW Set 10 Due
May 3-7	Hypothesis Testing 1-3	Definitions and Theory One Sample Hypothesis Tests Two Sample Hypothesis Tests	May 10: HW Set 11 Due
May 10-14	Analysis of Variance 1-3	Definitions and Theory Example Problems Post Hoc Testing	May 17: HW Set 12 Due
May 17-21	Regression 1-3	Definitions and Theory Workflow Example Problems	May 20: HW Set 13 Due

Keeping Up and Studying
This course is packed with information and our schedule will be tight. If you do R practice and a little homework every night you should be fine.

Academic Honesty:
Cheating and plagiarism will not be tolerated. If a student is suspected of either of these offenses he or she will be turned into the Provost. I encourage you to collaborate and help each other on projects, however the work you develop, and present must be yours. Upon being found guilty a grade of F will be recorded for the course.